BIT502 Fundamentals of Programming Assessment 3

Weighting

50%

Learning outcomes

2 Apply the principles of interface design to create usable and accessible user interfaces.
4 Design and develop a simple database application.

Instructions

Complete and submit your assessment according to the Open Polytechnic's Assessments webpage. This includes information on academic integrity, formatting, word limits and referencing.

- Include your name, student number and the assessment number.

Submission

- Submit your assessment in one zip file.
- Submit your work through your iQualify course.
- Emailed assessments will not be accepted.
- You will receive an automated notice advising you of your successful submission.

By submitting your assessment, you confirm that it is your own, original work.

© The Open Polytechnic of New Zealand

Introduction

In this assessment, you will design the wireframes and create forms for the interface of an application that interacts with a database. You will then use Python to create a functioning GUI based on those wireframes. You may need to consult the information available in Appendix A Membership Form and you will need to download Appendix B Database Information and Appendix C Database Script, which contain thedatabase information to complete this project.

Note: We recommend using Visual Studio Code instead of Jupyter Notebook as the code editor to complete this assessment.

Note: Do not use any additional libraries or dependencies outside of Tkinter and those used in the course material.

Generating the Database

In this assessment, you will be utilising a database for your application. You have been provided with the database structure in Appendix B Database Information, you should not change the structure of the database. Appendix C Database Script contains code to generate the database file, it also includes some example records you can use to insert some data into the database.

You may manually create the database, tables, and fields if you prefer, however, it is recommended you utilise the script provided to maintain consistency and reduce unexpected issues.

Please note, the script is only required to generate the database file. You will not need it once this is done and do not need to include it in your application or submission.

If you have trouble generating the database, you can use the pre-generated database file Appendix D Database Data.

Before you submit

- Wireframes and explanations can be submitted in a Word file or a PDF document that includes screenshots or images of the wireframes.
    - Include your course code (BIT502), name, student number, assessment number and page numbers on your Word document.
- Code files should be submitted together inside a zip file.
    - Rename the bookstore database file for your project.
    - Include your course code (BIT502), name, student number, and assessment number as a comment at the top of your code.
- Ensure your zip file includes:
    - your Python files and your database file.

After zipping your file, reopen it to make sure it works and contains the correct files.

Read the following case study and complete the tasks. Refer to the Marking Schedule forspecific information about marking.

© The Open Polytechnic of New Zealand

Case study

A local bookstore, The Aurora Archive, has been in business for several months and is seeing a steady increase in members. In addition to selling books, they have a few services that require the use of a membership subscription; these include:

- access to a private reading area
- a kid friendly environment that encourages reading becoming a lifelong habit
- various book and reading related activities.

The owners have asked you to develop some programs to help improve their administration service to save them time and reduce errors.

Simplifying the membership process with database access

The previous application has been quite successful for the store and managed to improve registration time. The convenience has been very helpful for the productivity of the store's staff.

Now they have had some time with the previous software you have built, the store ownershave found some other tasks that require a program to streamline their processes.

As the membership numbers have grown, they have found it difficult to find users in the text file. The file is getting larger, making it harder to work with. The ability to search for users based on different criteria would be very helpful for the business. Searching by name or other parameters would also be quite convenient for the staff.

Keeping track of the number of members in total and the most popular membership types is also challenging. Since all this information is digital, it would be convenient to have easy access to them.

After consideration, the best solution for storing and retrieving data will be with the use of a database, which will allow the staff to easily work with the client information.

You have been tasked with building a program that fulfils these needs. You will use your knowledge from the previous program to upgrade it and incorporate a database. The program should also provide the ability to search and gather statistics from this database.

While this program will include both staff and client functionality, you do not need to worry about any security or privacy issues at this time. A future update will take this into consideration.

© The Open Polytechnic of New Zealand

Task 1: Creating forms and wireframes

Your task is first to design the wireframes for the interface of an application that interacts with a database, following the principles for usable and accessible user interface. Then, you will use those wireframes to create forms or screens for the store membership application.

Please note that you need to create the wireframe design first before developing the actual screens or windows. This can be done either as a hand sketch or using any software tool. Do not use the actual GUI forms or screens as wireframes. The wireframe represents the initial phase of the design process, and once you are satisfied with it, you can proceed to design the actual forms and screens in Python.

Part A

1. Create wireframes for each screen/form using your medium of choice. You may create the wireframes digitally with software or on paper. Show all the UI elements you plan to include in the application and clearly identify how users will move between screens. The wireframes should be drawn separately from the application GUI.

   (5 marks)

2. For each screen or form, provide a brief explanation in a Word document about how you made your wireframe user-friendly and easy to access. Make sure to include screenshots or images of the wireframes for each screen or form in your explanation.

   (Word count guidance: 300 words)

   (5 marks)

Part B

Using Python and Tkinter, create five forms or screens for the bookstore membership application. Base the forms or screens on the wireframes you have designed for Part A and include all the necessary UI elements.

Note: Please refrain from using any libraries or dependencies beyond Tkinter.

Depending on your implementation, the application can be built using a single Python file or you may use multiple Python files with one for each form.

1. **Main screen:** This is a static window that should contain labels and navigation components/buttons to the rest of the forms, including the help window. This acts like a menu screen.

2. **Membership form:** This form is the same one created for Assessment 2. You can reuse the form without any changes to the UI or make some updates to improve it.

3. **Search form:** This form is used to search for members using the member's ID, last name, membership plan, or payment plan.

   Use appropriate UI components and layout to accommodate these search criteria. The form should also have a component to display the retrieved information. Ideally, multiple options should be selectable to filter results further.

4. **Statistics form:** The purpose of this form is to display various useful pieces of

© The Open Polytechnic of New Zealand

information about the members, such as the total number of members and which options users have selected, along with some information about expected income. This is an informative page for the owners to see which options are popular and have an idea of the business numbers for future decisions.

5. **Help screen:** This is a static window that should contain information on how to use this application. Briefly provide helpful instructions about the membership, search, and statistics forms.

(Part B: 20 marks)

(Task 1 total: 30 marks)

Task 2: Developing the application

After you have designed the wireframes and created the forms for the interface, your task is to design and develop the application functionality and connect the application to the database. You will need Appendix B Example Database Information, Appendix C Database Script and Appendix D Database Data to support your work.

You should not make modifications to the database structure.

Task 2 Parts B-D will involve interactions with the database. The first step is to correctly add the database to your project. After that, you need to make sure the application is functional and that all forms interact properly with the database.

Note: Regarding the database file, please follow these instructions:

- Place the database file in the same folder as your other Python source code files.
- When submitting your assessment, ensure that you include the database file along with your other code files.
- When defining the database file path in your code, do not use an absolute path (e.g., C:\temp). Instead, use a relative path.

Part A: Creating the main menu screen

The main menu screen allows users to navigate through the application. Using the GUI you created earlier, write the code for navigation to the five forms/screens:

- Main screen
- Membership form
- Search form
- Statistics form
- Help screen.

Make sure to allow the user to return to the menu from the other forms.

(10 marks)

Part B: Creating a membership form

Create a membership form that will store the data in the database provided. If you would

like, you can modify the form you made in Assessment 2, which saved the data to a file. For this, you will need to replace the code that writes to the text file with a new code that inserts the information into the database.

Ensure the form has all the functionality the user would expect, it should calculate and display the totals, check for issues, and consider how you can improve the user experience.

Each time the form is filled out, the application will need to insert the data as a new member in the database. The application should validate all user input and show a message when the data is successfully added or display an error message when there is a problem. The form should only be cleared when the user input is successful.

(20 marks)

Part C: Creating a search form

Create a search form that allows the user to search for members by the following:

- ID
- Name
- Membership plan (Standard / Premium / Kids)
- Payment plan (Monthly / Annual)

The search can return more than one result if multiple members are found. If there were no members found, display an appropriate message to the user. If possible, allow for multiple options to combine options and narrow down a search.

Make the functionality as user friendly as possible, such as ignoring case and allowing partial search.

Display the retrieved information in an appropriate format. Ideally, a table with one client per line.

(20 marks)

Part D: Creating a statistics form

Create a statistics form to help the staff access statistical information about the members. The form should display information about all members, including the following:

- Total number of members.
- Member count for each membership plan.
- Member count for monthly and annual.
- Member count that selected each optional extra.
- Member count that did not select any optional extras.
- Member count that has a library card.

The format to display the retrieved information is your choice. Make sure it is organized and easy to understand.

On the same form create a small table that displays the expected costs for the membership options, this will provide the client with a view of the income from the various options.

Note: The total income does not include any deductions from discounts.

© The Open Polytechnic of New Zealand

Option	Cost Per Unit	Member Amount	Total Income
Regular Plan			
Premium Plan			
Kids Plan			
Book Rental			
Private Area Access			
Monthly Booklet			
Online eBook Rental			

The "Cost Per Unit" is the price for that option, standard membership is $10 per month for example. Refer to the appendix for these values.

The "Member Amount" refers to how many members have selected this option. You will need to query the database for this value.

The "Total Income" will be calculated with your own method.

This table provides a simple approximation of monthly income, it does not need to apply discounts. Do not consider any discounts from library card discount or annual payment discount in this table.

(20 marks)

Quality control

Essential functions of your application

Your application must include the following.

- Registration of new members and ability to add them to the database.
- Ability to search for existing members in the database.
- Display overall information about the members.

Include a help section for the administration team on how to use this application.

Good programming practices

Your code should follow good programming practices, including:

- organising the code in a clear manner
- using meaningful variable naming
- minimising duplicate code
- using your own methods where sensible
- performing validation where necessary
- handling unexpected input or user mistakes appropriately
- ensuring the program does not crash or produce errors
- avoiding hard-coding values where possible

© The Open Polytechnic of New Zealand

- adding comments to explain the complex sections of the code.

Although well written code should not require much if any commenting, as the programs may undergo future development you have been asked to provide some comments to help explain any code that may not be obvious, such as calculations and formula in the code. You should not comment on every line.

Ensure your program is free from errors and compiles correctly.

(Task 2 total: 60 marks)

(Total: 100 marks)

© The Open Polytechnic of New Zealand

Marking Schedule

Task 1: Part A	4 – 5 marks	3.5 marks	2.5 – 3 marks	2 marks	1 – 1.5 marks
Creating forms and wireframes 1. Design and create five forms/screens for the application. • Main screen • Membership form • Search form • Statistics form • Help screen.	All five forms have been well designed in a way a UI designer could interpret the design without ambiguity with clear navigation. Wireframes are clear and readable.	All five forms have been designed and created; all are understandable and provide enough detail to comfortably build a GUI from. Navigation is clear and understandable. Wireframes have been created using another medium than screenshots of the final GUI.	All five forms have been designed and created; all are understandable and provide enough detail to roughly build a GUI from. The wireframes have not been designed by another medium and have used screenshots of the final GUI.	All five forms have been designed and created. The designs provide a rough guideline of the layout but are difficult to understand or lack enough detail to build the UI.	Fewer than five forms have been designed and created. Those forms that are created have errors and are not adequate to build the UI.
	4 – 5 marks	**3.5 marks**	**2.5 – 3 marks**	**2 marks**	**1 – 1.5 marks**
2. Explain how you have made your wireframe usable and accessible.	A logical explanation of how the wireframes are usable and accessible is provided. The explanation is justified with reference to other accessible and usable wireframes. The created wireframes are usable and accessible.	A logical explanation of how the wireframes are usable and accessible is provided. The created wireframes are usable and accessible.	A logical explanation of how the wireframes are usable and accessible is provided.	An explanation of how the wireframes are usable and accessible is provided; however, the explanation isinsufficient to prove the usability or accessibility of the wireframes.	An inaccurate explanation of how the wireframes are usable and accessible is provided.

© The Open Polytechnic of New Zealand

Task 1: Part B	17 – 20 marks	14 – 16 marks	10 – 13 marks	8 – 9 marks	1 – 7 marks
Create the five forms based on your wireframes, displaying all UI elements and allowing navigation between forms.	All forms are well laid out with good design and user experience. All functionality is accessible, and no errors exist with the interface or navigation.	Minor inaccuracies in the UIelements are present. All the forms have been created with the necessary elements. All forms perform as expected without any issues. Some minor inaccuracies or errors may exist.	All the forms have been created with the necessary elements. The forms are complete enough to provide the basic functionality. There may be some minor errors or challenges in the user experience but remain functional.	Most of the forms have been completed with some minor elements missing or a missing form.	One or more of the forms have been started but are mostly incomplete.
Task 2: Part A	**8 – 10 marks**	**6.5 – 7.5 marks**	**5 – 6 marks**	**4 – 4.5 marks**	**1 – 3.5 marks**
Developing the application Main menu screen	Navigation code has beenaccurately added for eachform. The menu is fully functional allowing navigation throughout the application. Good programming practices have been followed throughout.	Navigation code has been added for each form. There are one or two errors in the code leading to inaccurate navigation in a form. Good programming practices have mostly been followed.	Navigation code has been added for each form. Some functionality may be missing or have minor errors. Some good programming practices have been missed.	Navigation code has been added for each form. Some errors in code lead to inaccurate navigation. Many good programming practices have been missed.	Navigation code has not been added for all forms. Multiple errors in code lead to inaccurate navigation. Mostly poor programming practices have been followed.
Task 2: Part B	**16 – 20 marks**	**13 – 15.5 marks**	**10 – 12.5 marks**	**8 – 9.5 marks**	**1 – 7.5 marks**
Developing the application Membership form	The form performs all the expected functions without issues. Good programming practices have been followed throughout such as avoiding hard-coded values.	The form performs all the expected functions without issues, validation is performed, and some good programming practices are present.	Data is inserted correctly into the database and the basic functions are working. There are some minor issues or validation is missing.	The form has some functionality, but some problems exist or the data is not inserted into the database. Few good programming practices are present.	The form is present and can be navigated to but is not complete. Almost no good programming practices are present.

© The Open Polytechnic of New Zealand

Task 2: Part C	16 – 20 marks	13 – 15.5 marks	10 – 12.5 marks	8 – 9.5 marks	1 – 7.5 marks
Developing the application Search form code	Search is fully functional allowing the user to combine or mix options to retrieve the expected results. User experience is done well. Good programming practices have been followed throughout, such as using methods where sensible.	Search is fully functional allowing the user to combine or mix options to retrieve the expected results. The search form is easy to use without any challenge. Mostly good programming practices are present.	Each of the 4 search options are functioning, returning accurate results. Combining search options have not yet been added. The basic functionality has been completed but there may be challenges with the user experience, such as not displaying a warning if no results found or other minor issues. Some good programming practices are present.	Some of the search functions are working, allowing the user to perform a search and obtain a result. Some options are not complete or results are not accurate. Few good programming practices are present.	Not all data is retrievable using the search mechanism. There are significant errors in data displayed in search results. Search functionality has been started but there are some significant issues where results may not be accurate. Almost no good programming practices are present.
Task 2: Part D	**16 – 20 marks**	**13 – 15.5 marks**	**10 – 12.5 marks**	**8 – 9.5 marks**	**1 – 7.5 marks**
Developing the application Statistics form code	Data is retrieved, calculated correctly and displayed in an appropriate format. Form is completely usable. Good programming practices have been followed throughout.	Data is retrieved and displayed for the expected output. There are no issues with the results. Some error checking has been performed. Mostly good programming practices are present.	Data is retrieved and displayed for the expected output. Some minor issues may exist in the calculations or output. Some good programming practices are present.	Data is retrieved correctly and displayed, but not all information is displayed correctly. Few good programming practices are present.	The form has been created and some functionality has been attempted but there are significant errors or missing functionality. Almost no good programming practices are present.

© The Open Polytechnic of New Zealand